INCH-POUND

A-A-59761
14 December 2004

COMMERCIAL ITEM DESCRIPTION

GENERATOR, SIGNAL
(1.8 GHz to 4.5 GHz)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) describes a solid state, general purpose, 1.8 GHz to 4.5 GHz, signal generator. This CID is meant as a minimum requirement for the signal generator in which only those manufacturers that meet or surpass the following requirements are qualified per this CID.

2. SALIENT CHARACTERISTICS. The equipment shall be capable of operation and product sample testing, if so requested, within the accuracies, limits, and specifications herein.

2.1 Classification. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be Class 3 in accordance with MIL-PRF-28800, except as specified herein.

2.2 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

2.3 Temperature Operating. The equipment shall conform to its specified performance and accuracy while being operated as a volt meter in a temperature range of 0 °C to 50 °C.

2.4 Temperature Non-Operating. The equipment shall conform to its specified performance and accuracy after being stored in an environment in the temperature range of -40 °C to 71 °C.

2.5 Electrical Power Sources. The equipment shall operate from nominal single-phase commercial, military, and shipboard power sources of 110 and 220 volts AC (± 10%) at line frequencies of 48 Hz to 63 Hz.

Comments, suggestions, or questions on this document should be addressed to: WR-ALC/LEEC 295 Byron Street, Robins AFB, GA 31098-1611. Since contact information can change; you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil

FSC 6625

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2.6 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 4,000 hours of operation with a statistical certainty of 95%.

2.7 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater, based on an operating time of 2000 hours per year.

2.8 Performance Requirements. The instrument shall meet all requirements specified herein, after the warm-up period specified in MIL-PRF-28800, under all combinations of input power conditions, output signal conditions, and operating service condition as specified in MIL-PRF-28800.

2.9 Frequency Range. The signal generator shall have a frequency range of 1.8 GHz to 4.5 GHz.

2.10 Resolution. The generator shall have a frequency resolution of 1 Hz.

2.11 Accuracy. The frequency accuracy shall be based on the internal reference oscillator aging rate, which shall be 1 x 10^{-10} /day, 1 x 10^{-9} /month

2.12 RF Output. The signal generator range shall be a minimum output range of

1800 MHz to 3000 MHz : +10 dBm to –127 dBm
> 3000 MHz : +3 dBm to –127 dBm

2.13 RF Output Power Accuracy The output power accuracy shall be

3000 to 4500 MHz: +3 dBm to -127 dBm

2.2 Spectral Purity.

2.2.1 Harmonics. The instrument shall have ≤ -25dBc harmonics over the entire frequency range with output ≤ +4 dBm.

2.2.2 Sub-Harmonics. The instrument shall have sub-harmonics shall be as follows with output ≤ +4 dBm:

< 2.3 GHz: < -40 dBc
≤ 4.5 GHz: < -30 dBc

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2.2.3 Non-Harmonics. The instrument shall have non-harmonics levels of at an output of ≤ +7 dBm and > 3 kHz offset.

≤ 2 GHz: < -70 dBc
2 GHz to 4 GHz: > -68 dBc
≥ 4 GHz: < -62dbc

2.3 Pulse Modulation.

2.3.1 On/Off Ratio. The on/off ratio shall be at least 20 dB for square wave and pulse modulation.

2.3.2 Rise and Fall Time. The rise/fall time shall be 2 μs.

2.4 Amplitude Modulation (AM).

2.4.1 AM Rate. The AM bandwidth at 1 db shall be DC to 30 kHz.
2.4.2 AM Depth. The AM depth shall be 0 to 80% typical (up to 100kHz).

2.5 Frequency Modulation (FM).

2.5.1 FM Modulation. The signal generator shall provide any waveshape frequency between 10 Hz and 0.5 MHz.

2.6 Weight. The total weight of the equipment, excluding accessories and manuals, shall not exceed a two-person lift of 74 pounds, (33.6 kilograms).

2.7 Dimensions. The width and depth of the equipment shall be compatible with mounting in an ANSI/EIA-310-D standard equipment rack. The height shall be compatible with four standard height units, 4U. One unit or "U" is defined by ANSI/EIA-310-D to be 1.75 inches or 44.45 mm.

2.8 Rack Mount. The analyzer shall be capable of being rack-mounted in a standard ANSI/EIA-310-D equipment rack and include a rack-mount conversion kit if needed.

2.9 Manuals. The equipment shall be delivered with operator, maintenance/service, illustrated parts, and calibration manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800, shall be as specified in the contract or order.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5. PACKAGING.

5.1 Preservation, Packing, and Marking. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

6.2 Sources of documents.

6.2.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained at http://assist.daps.dla.mil or available from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.2.2 The Code of Federal Regulations (CFR) may be obtained at http://www.gpoaccess.gov/cfr/index.html or available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.2.3 ANSI/EIA standards can be obtained at http://www.ansi.org or http://www.eia.org or available from the Electronics Industry Association, Engineering Department, 2001 Pennsylvania Ave., N.W., Washington, D.C., 20006. Phone: 1-800-854-7179 (USA and Canada).

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Military Interest

Custodian:
Air Force 99

Preparing Activity:
Air Force - 84

Agent:
Air Force - 99

(Project 6625-F103)

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